



In Pursuit of Excellence

STUDENT MODULE GUIDE

DES15W0 00/2026
DEV15W0 00/2026

DEVELOPMENT SOFTWARE 1

FACULTY OF ENGINEERING, BUILT ENVIRONMENT & INFORMATION
TECHNOLOGY

DEPARTMENT : DEPARTMENT OF INFORMATION AND COMMUNICATION SYSTEMS
CAMPUS : BUFFALO CITY CAMPUS

IMPORTANT INFORMATION

NB: The purpose of this document is to provide you with the necessary guidelines for the successful completion of this module in terms of the theoretical and the practical aspects of its teaching, its learning, and its assessment. This document also contains relevant information on the administration of this module including the rules of engagement on your communication with your peers as well as with the Department's staff.



Excellence In Teaching & Learning

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1. Introduction & Welcome

On behalf of Walter Sisulu University and the Faculty, we would like to welcome you to the Information Technology programme, particularly to Development Software 1. Computing skills play an increasingly vital role in almost all occupations, and contribute to productivity gains, innovation, and employment growth across all South African and international industries. This is leading to exciting career opportunities. Many new jobs have been created in the past decade, and the long-term career prospects continue to strengthen. The academic focus of the Department is to offer programmes specifically aimed at preparing students for the workplace by providing training that enables diplomates and graduates to demonstrate competence, knowledge, and judgment in addressing discipline-related problems.

Please consult the Faculty Prospectus for a description of the different specialisations and information about all of the courses offered by the Information Technology Department at each campus. The prospectus will also show you how the courses must be combined in order to follow a particular specialisation.

Development Software 1 is one of the four compulsory subjects that must be completed in the common first year / extended curriculum first and second years of all diploma specialisations. It is an exciting and stimulating course which introduces you to the world of computer programming. Successful completion of the course will enable you to pursue a potential career as a software developer or provide you with the necessary background skills to follow one of the other specialisations on offer within our department.

The purpose of this Student Module Guide is to provide you with written guidelines as to what to expect of the University and what the University in turn expects of you, with regard to this course. You are given administrative details of the Faculty and Department as well as the contact details of the lecturers teaching this course at different delivery sites. The general course information shows you the number of lectures, practical and tutorial periods a week and provides information about delivery mode, online learning and additional time you will be expected to spend on course-related work. The responsibilities of the students with regard to participation and attendance are stated. Details of all resources available as well as student support services are provided. Full technical details, including the purpose of the qualification, entry requirements, NQF level, credits, and exit level outcomes, are provided. Comprehensive information about the assessments approach used in the course are included. Details about the topics and specific outcomes of the Units in the Module are also given, as well as a proposed work plan for the year.

2. Module Purpose & Outcomes

2.1 Module Purpose

The course is designed to equip students with a solid understanding of the fundamental principles of programming, which are applicable to all computer programming languages. Students are taught how to analyse problems and how to write structured algorithmic solutions using (formalised) pseudocode and the C # programming language. Students are introduced to the concept of the Program Development Cycle and taught how to use tools associated with each stage.

The same course content is offered in the Mainstream and Extended Curriculum programmes. Students have additional contact time in the extended programme.

Mainstream: subject codes DES15W0

Development Software 1 is one of the four compulsory subjects that must be completed in the common first year of any of the mainstream IT Diploma specialisations.

The other subjects are **Information Systems 1**, **Systems Software 1**, and **IT Skills 1**.

The contents of the course are supplemented by the contents of the other courses.

Extended Curriculum: subject codes DEV15W0

Development Software 1 is one of the four compulsory subjects that must be completed in the common second year of any of the ECP IT Diploma specialisations.

The other subjects are **Information Systems 1B**, **Systems Software 1B**, and **IT Skills 1 B**.

The contents of the course are supplemented by the contents of the other courses.

2.2 Module Learning Outcomes and Assessment Criteria

Module Learning Outcomes	Module Assessment Criteria
The student will be able to: 1. Develop and practice the problem-solving skills and logical approach needed to analyse a programming problem and design and test a solution using internationally recognised tools and techniques associated with the Program Development Cycle, specifically IPO charts, pseudocode, flowcharts, and trace tables.	1. The role and major components of software development within digital computing can be explained. 2. Each of the major steps in the Program Development Cycle can be identified and explained. 3. The use of variables to store, process, and retrieve data can be applied in solutions. 4. IPO charts can be used to assist with the analysis stage of the cycle to identify
	the input, processing, and output components of a given problem. 5. The syntax rules and use of formalised pseudocode and correct logic can be applied for the design and writing of algorithms as solutions to problems. 6. The rules and use of formalised flowcharts and correct logic can be applied for the graphical representation of algorithms as solutions to problems. 7. The rules and use of formalised trace tables can be used to desk-check the logic in algorithms and flowcharts.

© Walter Sisulu University Student Module Guide	The student will be able to:	1. The six basic computer operations are
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	2. Design, trace, and write solutions in	3. The correct formalized pseudocode
--	--	--------------------------------------

	structured (formalised) pseudocode using	language instructions and variables can
--	--	---

input from a keyboard, output to a screen,
--

be written to receive data and store

	variables, arithmetic, relational, and logical	results of calculations and output
--	--	------------------------------------

operators, and the basic control structures,	results.
--	----------

	loop, and DO WHILE pre-test loop, Nested IF,	correct	formalized	pseudocode
--	--	---------	------------	------------

and Select Case.	instructions can be written to illustrate
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		the use of the sequence control
--	--	---------------------------------

		formalized pseudocode instructions and
--	--	--

		language rules can be written to use the
--	--	--

		structures - utilizing the correct
--	--	------------------------------------

		5. An algorithm using the correct
--	--	-----------------------------------

		formalized pseudocode instructions and
--	--	--

		language rules can be written to use the
--	--	--

	6. If appropriate in a solution, the correct formalized pseudocode instructions and language rules can be written to use the Select Case and Nested IF selection control structures - utilizing the correct relational and logical operators.
The student will be able to: 3. Code, test, or run and debug programs using the C # programming language using input from a keyboard, output to a screen, variables, arithmetic, relational and logical operators, and the basic control structures, including Simple and Compound IF, FOR loop, and DO WHILE pre-test loop, Nested IF, and Select Case.	1. The structure and processing of the C # programming language are understood. 2. The C # Interactive Development Environment can be used to create, save, execute, test, and debug C # programs on a computer. 3. The correct C # language instructions and variables can be written to receive data, store the results of calculations, and output results. 4. A logically correct program using the correct formalized C # instructions can be written to illustrate the use of the sequence control structure. 5. If needed in a solution, the correct C # instructions and language rules can be written to use the Simple and Compound selection control structures - utilizing the correct relational and logical operators. 6. A program using the correct C # instructions and language rules can be written to use the FOR and DO WHILE repetition control structures when needed in solutions. 7. If appropriate in a solution, the correct C # instructions and language rules can be written to use the Select Case and Nested IF selection control structures - utilizing the correct relational and logical operators.

2.3 Module NQF Level

6

2.4 Pre - and co-requisites

Refer to the faculty prospectus for the minimum entry-level requirements for all IT Diplomas.

Extended Curriculum (ECP): for subject codes DEV15W0

- Information Technology Skills 1A PLUS System Software 1A

2.5 Module Credits

30 credits

2.6 Notional Hours Required

300

3. Qualification Information

3.1 Qualification Rules

Each diploma (all specialisations, including Extended Curriculum) is a 360-credit qualification. To graduate, a student must pass each of the compulsory modules listed in the prospectus for the qualification.

Year 1 courses are common to all qualifications, and a pass in each is required to graduate.

3.2 Qualification Title

Mainstream: Subject code DES15W0

Diploma in ICT in Application Development **(W6009)**

Diploma in ICT in Business Analysis **(W60011)**

Extended Curriculum (ECP): Subject code DEV15W0

Diploma in ICT in Application Development ECP **(W60010)**

Diploma in ICT in Business Analysis ECP **(W60012)**

3.3 Qualification NQF Level

6

3.4 Qualification Total Credits

360 credits

4. Mode of Delivery & Access to Online Systems

Formal Contact Time – attendance of all classes is compulsory

Mainstream: Subject codes DES15W0

Theory	3 x 45-minute periods per week	
Practical (lecturer-led lab)	4 x 45-minute periods per week	Extended

Curriculum (ECP): Subject codes DEV15W0

Theory	4 x 45-minute periods per week
Tutorial	4 x 45-minute periods per week
Practical (lecturer-led lab)	4 x 45-minute periods per week

This module is presented using a blended learning approach. In other words, it is presented both face-to-face on campus in the classroom/lab and online using MS Teams and the Wiseup (Moodle) platform.

Face-to-face (classroom/lab) delivery

The Practical component will be delivered on campus.

All students will be allocated to a group and will be given a timetable of the weekly practical sessions they are required to attend. Evidence of homework preparation is expected for entry to the practical sessions.

Access to online systems

Registered students will be provided with logins, passwords, and links to access the MS Teams and WiseUP platforms. Students will also be given personal email addresses on the WSU student email system, MS Outlook 365, as well as logins for access to the WSU Student Portal.

Devices for online systems

Students are given an opportunity to purchase laptops.

In some residences, Wi-Fi is provided, but if it is not available, students should use their allocated data for their compulsory online tasks. Additionally, Development Software 1 is a practical course that requires students to write programs on PCs or laptops; therefore, students are strongly advised to apply for a laptop in their first year.

Problems with online access, device

Students will be given the contact details of the relevant campus support staff to consult for various types of problems they may encounter.

Non-Contact Time

Students are expected to spend at least 5 hours a week completing work outside of formal contact time and attending PAL/FACE sessions. This will involve completing homework and preparing for the next sessions, as well as utilizing the online learning platform.

5. Contact with the Lecturers, Department, and Faculty

5.1 Lecturers

Lecturer	Delivery Site	Office No.	Telephone Email	Consultation Times
Ms. S Twetwa –Dube	Potsdam	PALG002	043 708 5296 stwetwa@wsu.ac.za	To be advised
Mrs. S Vijayalekshmi	Potsdam	PBF025	0437085201 svijayalekshmi@wsu.ac.za	To be advised

WhatsApp, WSU email, and WSU WiseUp

All students will be added to the class WhatsApp group, which will be managed by the Class Representative. All day-to-day communication within the group will be conducted via WhatsApp. Additional information will be communicated to individuals or the group via the WSU email system (MS Outlook 365) and the various online systems in use (e.g., Moodle and MS Teams).

5.2 Department

		Delivery Site	Office No.	Telephone Email / Fax
Head of Department	Dr J Lukose	Potsdam	PBF039	043 708 5405 jlukose@wsu.ac.za
Administrator	Ms. U MtshizanaNgandu	Potsdam	PBF032	043 708 5405 umtshizana@wsu.ac.za
Admin Assistant	Ms. S Finini	Potsdam	PBF032	043 708 5405 sfinini@wsu.ac.za

5.3 Faculty

		Delivery Site	Office No.	Telephone Email / Fax
Executive Dean	Prof W.D Thwala	Chiselhurst	F49	043 709 4069 wthwala@wsu.ac.za
Faculty Officer	Ms. G Lindani	Chiselhurst	F44	043 709 4004 glindani@wsu.ac.za
Faculty Secretary	Ms. B Panyaza	Chiselhurst	F49	043 709 4069 bpanyaza@wsu.ac.za

The university often communicates with the student body using the WSU email system or the WSU Facebook page (see Other Resources below).

6. Resources

6.1 Study Material

All students are provided with a MODULE STUDY AID – i.e., a comprehensive set of notes developed specifically for the course as well as the pack of exercises they will complete during the year. These are provided in hard copy and online in Moodle.

All slides and video recordings of any online Teams sessions are provided in WiseUp in addition to other resources.

6.2 Prescribed and Recommended Material

Prescribed Textbook: The purchase of this textbook is compulsory:

Pretorius C. M. and Erasmus H. G. (2012). *Basic Programming Principles*. Second Edition. Cape Town: Pearson Education. ISBN: 9781775786030

Programming Basics with C# by Svetlin Nakov & Team (ISBN: 978-619-00-0902-3), published by Faber Publishing in Sofia, 2019, is licensed under CC-BY-SA.

Other useful references (see WiseUp for additional resources)

Bradley R. (1988). *Understanding Computer Science for Advanced Level*. London: Century Hutchinson Ltd.

Farrell, J. (2002). *Programming Logic and Design, Comprehensive, Second Edition*. Canada: Course Technology, a division of Thomson Learning, Inc.

Robertson, L. (2004). *Simple Program Design: A Step-by-step Approach, Fourth Edition*. Hong Kong: Course Technology, a division of Thomson Learning, Inc.

Sprankle, M. (2005). *Problem solving and Programming concepts, Seventh Edition*. Prentice Hall Inc.

<https://github.com/SoftUni/Programming-Basics-Book-CSharp-EN>

Some useful web sites (valid as at 01/02/2020)

General: <http://www.google.co.za> <http://en.wikipedia.org> <http://dictionary.reference.com>

Google courses: <https://learndigital.withgoogle.com/digitalskills/courses> CISCO courses:

<https://www.netacad.com/courses/all-courses>

<https://www.netacad.com/careers/pathways-and-certifications> IBM

courses:

<https://www.ibm.com/services/learning/ites.wss/zaen?pageType=page&c=a0014719>

MICROSOFT courses : <https://www.microsoft.com/en-us/learning/training.aspx>

<https://www.edx.org/school/microsoft>

Dept of Higher Education and Training: <http://www.education.gov.za>

Centre for Higher Education: <http://www.che.org.za>

MICT SETA: <http://www.mict.org.za>

https://youtu.be/_F606F3OgmQ

6.3 Electronic Resources / Material

All course notes, slides, recordings, etc., will be posted on the Learner Moodle Management System.

Important communication about homework, assignments, tests etc. will also be posted online and shared on the class WhatsApp groups.

6.4 Other Resources

WSU General prospectus:

<https://www.wsu.ac.za/index.php/study-with-us/quick-links/prospectus/general-prospectus>

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<https://www.wsu.ac.za/index.php/study-with-us/quick-links/prospectus/buffalo-city>

WSU Facebook page:

<https://www.facebook.com/OfficialWSU>

WSU website: <https://www.wsu.ac.za/>

Others resources will be shared mainly on the WiseUp system.

6.5 Accessing the Library

The University provides a wide variety of services through the library at each campus. Students should visit the library at their campus and familiarise themselves with the services available.

The library also provides comprehensive online services and access to online publications, text books, journals etc.

For more information, see Section 7 in the WSU General Prospectus.

7. Student Support Services

7.1 Student Counseling

The University is committed to providing students with a holistic learning experience through the Department of Student Affairs, which manages student enquiries, complaints, and student life in general on a daily basis. Utilizing the services of dedicated and competent staff, the Department of Student Affairs offers students relevant training and support in areas such as counselling, health, personal development, and more. The Department strives to create and sustain a vibrant and well-resourced learning environment in which respectful relationships are fostered and maintained.

Students can refer to section 13 of the WSU General Prospectus for further details.

7.2 Tutor & PAL / FACES Services

Peer Assisted Learning (PAL) or Foundation Academic & Career Enhancers (FACES) – attendance is compulsory

All Development Software 1 students are expected to attend at least 60 minutes of PAL/FACE sessions per week, outside of the formal contact time.

8. Lecture Room Expectations

According to university policy, all face-to-face and online classes must be attended punctually and regularly.

The department has adopted the following attendance policy:

- a. If a student does not attend 80% of classes between tests, the next test will not be marked, meaning the student will be awarded 0% for the next test.
- b. If a student does not attend 80% of classes during the year, the student will not be admitted to the fourth and supplementary tests.

Registers will be taken at all classes and filed for future reference.

Students who are unable to attend a formal contact session must provide the lecturer with a written note explaining their non-attendance and, if applicable, a medical certificate within five working days. These notes will be kept in the course file to check student attendance.

Development Software 1 is recognised as a challenging course and is included in the Peer Assisted Learning programme (PAL) & Foundation Academic & Career Enhancers (FACES) coordinated by the WSU – Directorate for Learning and Teaching (DLT). This programme is run for the benefit of all students, more especially those who are struggling with the course content. Students are expected to attend as many PAL sessions as possible each week. Registers will be taken at all PAL/FACE sessions and filed for future reference.

Students are also expected to spend a minimum of 5 hours a week completing homework exercises, revising, preparing for classes and tests, and participating in WiseUp online activities, such as group discussions. Research has shown that attending to these tasks together with classmates is generally beneficial to all students.

Misunderstandings about course content or failure to grasp a topic can often be clarified in one-on-one sessions with the lecturer. Students are encouraged to consult their lecturers if they have any questions or concerns. It is also important for a lecturer to be aware of any personal problems that may affect a student's performance. If necessary, the lecturer will be able to direct

the student to the various support services offered by the University, such as Student Counselling Services, Health Services, and HIV/AIDS Support Services, among others.

Students are requested to make appointments to meet with lecturers during the available consultation times. As this will not always be possible, however, lecturers are also expected to make time to meet with students in 'emergency' situations.

Lecturers are expected to attend all scheduled contact classes. If possible, notice will be given if a lecturer is unable to attend a class, and arrangements will be made by the lecturer to cover the missed work.

Students are advised to check the department notice board, WiseUP announcements, or with their class reps on a daily basis for any information pertaining to their course, classes, etc.

Class Representatives, WhatsApp Class groups, Email, and Complaint Procedures

It is university policy that each class should elect two class representatives at the beginning of each year. One of the roles of the class representatives is to convey information from the lecturer/department to other students - this is usually done using WhatsApp groups. Students who do not have access to WhatsApp must arrange to keep up to date through classmates. The class representatives may also be called upon to represent the students in meetings with Department staff and/or management.

WSU student email is also used by lecturers and various university divisions and departments to engage with the student body and on an individual basis with students. Students should check their email daily and respond as/when requested.

The correct procedure for any student who has a complaint about a lecturer or course is for the student to first approach the lecturer and attempt to resolve the issue. If this is unsuccessful, the student should complete a student complaint form (available from the secretary) and take the issue to the Head of Department. If many students in a class have a problem with a lecturer or course, the problem must be detailed on a student complaint form, and the form should be passed to the class representatives, who should act on behalf of the students by first approaching the lecturer and then, if necessary, escalating the problem to the HOD.

9. Rules for Online Engagement (Netiquette)

The following set of rules is meant to guide our online Teams engagement:

1. Every learner has to log on to the learning platform so their presence is noted in the register.
 2. Students are encouraged to participate actively throughout each session. Questions can be asked by raising hands on the platform. Students are expected to post pictures of required homework in the Chat window and participate in discussions using the Chat window.
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3. Students are encouraged to log on to the portal at least 2 minutes before the start of each class.
4. Videos of each online lesson shall be made available, but attendance at live sessions is compulsory.
5. Always mute your microphone and only unmute it when asked to do so or when making a contribution.
6. Video mode must be kept off at all times to ensure the quality of audio during lessons.
7. Never interject a speaker without following due procedures.

10. Formative Assessment

This type of assessment will be used to inform both the student and the lecturer and will not be allocated a mark. It will allow the student and lecturer to monitor his/her progress in achieving the course goals and specific outcomes. It will take the form of self-review, peer review, lecturer review, online quizzes, and other assessments, and the student should view them as valuable learning aids. It will also provide valuable feedback to the lecturer regarding his/her instruction techniques, topics that require revision or additional attention, etc. Students will be asked to provide comprehensive evaluations of the course and lecturer at least once during the course cycle.

11. Summative Assessment

This assessment is used to formally measure the learning of particular skills, knowledge, and understanding, and is associated with a recorded mark.

Development Software 1 is a year course that is exam-based.

NB: All summative assessments will count towards the Duly Performed (DP) mark (also called year mark). All summative tests will be administered in an INVIGILATED environment ON CAMPUS – whether the assessments are online in a lab or on paper in a classroom.

Composition of the DP mark

The DP mark for Development Software 1 will be calculated from summative tests (4 are proposed) using the weightings given in 11.1.

Examination

There will be a three-hour written examination at the end of the year. According to University policy, entrance into the examination is subject to a minimum DP mark of 40%. The minimum mark required to pass an examination is 40%. If a learner score below 40% in the examination, he/she does not qualify for a final mark and does not pass the course.

Final Mark (before supplementary examination) The Final

Mark is calculated as follows:

DP	40%
Examination	60%
Total:	100%

A learner must score at least 50% for the final mark (calculated as shown above) in order to pass the course. A final mark in the range 40-49% qualifies a learner for a supplementary examination.

Supplementary Examination

There will be a three-hour written examination at the end of the year.

According to University policy, entrance into the supplementary examination is subject to a final mark in the range 40-49%.

If a learner scores below 40% in the supplementary examination, he/she do not pass the course.

Final Mark (after supplementary examination).

The Final Mark is calculated as follows:

DP	40%
Supplementary Examination	60%
Total:	100%

A learner must score at least 50% for the final mark (calculated as shown above) in order to pass the module. For requests regarding remarks, absences from examinations due to illness, special examinations, and other related matters, please refer to the General Prospectus for the relevant rules.

11.1 Tests

There will be 4 (four) main tests during the year, each of which will be assigned a weighting as shown below:

TEST 1- 15% (Assessment 1 & 2), TEST 2- 20%, TEST 3 - 30% (Assessment 1 & 2), TEST 4- 35%

NOTE: Students may not miss more than one of the first 3 tests.

11.2 Final mark

The sum of the weightings applied to each of the 4 test marks will make up the final mark for the course at the end of the year, as shown below:

FINAL MARK = (0.4 x DP i.e., year mark) + (0.6 x exam mark)

A student must score at least 50% for the final mark to pass the course. A final mark below 50% qualifies a student for a supplementary.

11.3 Progression

A final mark below 50% means the student has failed the course and must repeat it.

11.4 Missed Assessments

Medical certificates regarding missed tests/or assignments, and other relevant events must be submitted to the lecturer, administrative assistant, or department secretary within five working days of the test date or due date. Medical certificates not received within 5 days will not be considered – unless the student was hospitalized.

Students will generally not be permitted to submit more than one medical certificate for a formal test. If such a case arises, a special arrangement must be negotiated with the lecturer and HOD.

11.5 Assessment Criteria

Each summative assessment during the year will assess how well the student has achieved the outcomes of specific Units. Students will be told which Units (and outcomes) will be included prior to the assessment. The main exam and the supplementary test will assess how well the student has achieved the exit-level outcomes of the course, as well as the specific outcomes of all units.

11.6 Moderation

All summative assessment question papers and memoranda for Development Software 1 are internally moderated during the year.

12. Assessments: Semester 1 – First Semester Compulsory Assessments

12.1 Assessment 01 – Semester 1

12.1.1 Instructions & Due Date

Test 1: This closed-book test (comprising two assessments) will be administered on campus, online in a lab, or on paper. The format will be a mixture of multiple-choice, True/False, and Essay questions, in which students will be required to design and write solutions for given problems.

These 2 tests will count towards 15% of the year mark (DP) – see 11.2 above.

The scope of the test will depend on the work covered (see item 15 below) and will be shared with students at least seven days prior to the test date.

The date will be approximately in the 9th teaching week.

12.1.2 Assessment Questions

These will be released on the day of the test only.

12.2 Assessment 02 – Semester 1

12.2.1 Instructions & Due Date

Test 2: This closed-book test will take place on campus, online in a lab, or on paper. The format will be a mixture of multiple-choice, True/False, and Essay questions, in which students will be required to design and write solutions for given problems. The test will count towards 20% of the year mark (DP) – see 11.2 above. The scope of the test will depend on the work covered (see item 15 below) and will be shared with students at least seven days prior to the test date. The date will be approximately in the 13th teaching week.

12.2.2 Assessment Questions

These will be released on the day of the test only.

13. Assessments: Semester 2 – Second Semester Compulsory Assessments

13.1 Assessment 03 – Semester 2

13.1.1 Instructions & Due Date

Test 3: This closed-book test (comprising two assessments) will be administered on campus, online in a lab, or on paper.

The format will be a mixture of multiple-choice, True/False, and Essay questions, in which students will be required to design and write solutions for given problems.

These 2 tests will count towards 30% of the year mark (DP) – see 11.2 above. The scope of the test will depend on the work covered (see item 15 below) and will be shared with students at least seven days prior to the test date.

The date will be approximately in the 20th teaching week.

13.1.2 Assessment Questions

These will be released on the day of the test only.

13.2 Assessment 04 – Semester 2

13.2.1 Instructions & Due Date

Test 4: This closed-book test will take place on campus, online in a lab, or on paper. The format will be a mixture of multiple-choice, True/False, and Essay questions, in which students will be required to design and write solutions for given problems. The test will count towards 35% of the year mark (DP) – see 11.2 above. The scope of the test will encompass all work covered during the year (see item 15 below) and will be shared with students at least seven days prior to the test date. The date will be approximately in the 28th teaching week.

13.2.2 Assessment Questions

These will be released on the day of the test only.

14. Module Content

UNIT NUMBER and TOPIC	OUTCOMES and ASSESSMENT CRITERIA
UNIT 1: THE SOFTWARE DEVELOPMENT ENVIRONMENT: INTRODUCTION TO COMPUTERS, SYSTEMS AND PROGRAMMING	a) Identify the computer as a tool and relate it to other tools they are familiar with b) Explain the concept of a system c) Distinguish the major subsystems of a computer system and describe them briefly d) Describe and explain the major phases of a computer software application e) Explain binary digits and their relationship to the architecture of a digital computer f) List and explain the various generations in the development of programming languages g) List and explain the major stages in the program development cycle
UNIT 2: GENERAL PROBLEM SOLVING	a) List and describe the five steps in the problem-solving approach b) Write English algorithms for solving simple problems encountered in everyday life
UNIT 3: DATA PROCESSING: DATA TYPES AND STORAGE, VARIABLES, BASIC OPERATIONS, PSEUDOCODE	a) Distinguish between the most important data types b) Explain how different data types are represented in a computer and stored in memory c) Understand the use of variables in programming d) List the 6 basic operations a computer can perform e) Understand the use of pseudocode for problem-solving f) Use the correct syntax and rules for pseudocode instructions to: get data from a keyboard, display output to a screen, assign data to a variable, perform calculations using the arithmetic operators
UNIT 4: PROBLEM-SOLVING TOOLS: IPO CHART, FLOWCHART, TRACE TABLE	a) Use an IPO Chart as an analysis tool to represent the logic of a problem b) Use a flowchart as a design tool to solve a problem c) Draw a trace table to test the logic in a flowchart and in a pseudocode algorithm

UNIT 5:

a) Describe the three basic control structures

	b) Develop an algorithm using simple pseudocode and the sequence
--	--

CONTROL	control structure (using input from a keyboard and displaying output
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STRUCTURES:

on a screen)

SEQUENCE,	c) Describe the purpose and use of the selection control structure
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COMPOUND IF,	e) Describe and use a compound IF selection control structure using
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CASE)	f) Combine and evaluate expressions using arithmetic, relational, and
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ITERATION	logical operators according to the rules of precedence
-----------	--

	selection
--	-----------

	h) Correctly show simple and compound selections in flowcharts and in
--	---

	trace tables
--	--------------

	i) Describe the purpose and use of the iteration control structure
--	--

	j) Describe and use a FOR iteration control structure in an algorithm
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	k) Describe and use a pre-test DO WHILE iteration control structure with
--	--

	a sentinel value in an algorithm
--	----------------------------------

	l) Correctly show FOR and DO WHILE iteration in trace tables
--	--

	m) Describe and use a Nested IF selection control structure in an
--	---

	algorithm
--	-----------

	n) Describe and use a Select Case selection control structure in an
--	---

	algorithm
--	-----------

	o) Correctly show advanced selection structures in trace tables
--	---

	p) Design pseudocode and C # programs using all of the control
--	--

	structures
--	------------

UNIT 6: C # Programming	a) Use the basic C # IDE b) Describe the various files that make up a C # project c) Create, save, and re-open a simple C # project with the correct name in a specified folder d) Use the correct C # instructions to write program statements for all of the instructions used in pseudocode
	e) Understand the difference between design time, break time, and run time f) Understand the importance of testing a program and be able to identify the differences between compilation, runtime, and logic errors g) Use the C # debug tool to trace the flow of program execution and examine the contents of variables

15. Scheme of Work (Module Timetable)

TEACHING WEEK #	END DATE	UNITS	TOPICS	PRACTICAL (EXERCISES)
1		<u>INTRO DUCTION</u>	Work through SMG, Introduce selves to the group	
2		<u>UNIT 1: PART 1</u>	<u>Unit 1: Intro to computers, systems</u> a. computers & systems (Unit 1: 1.1, 1.2, 1.3,1.4)	
3		<u>UNIT 1: PART 2</u>	<u>Unit 1: Intro to Programming</u> a. languages, PDC (Unit 1: 1.5)	
4		<u>UNIT 2</u>	<u>Unit 2: General Problem Solving</u>	<u>Exercises 01</u>
5		<u>UNIT 3: PART 1</u>	<u>Unit 3: Data Processing: General Concepts</u> a. Intro & overview b. Data representation in a computer (Unit 3.1) d. 6 basic operations (3.2) e. Variables ** naming & data type rules ** (Unit 3.2)	<u>Exercises on the slides</u>
6		<u>UNIT 3: PART 2</u>	<u>Unit 3: 3.2 Pseudocode</u> a) Pseudocode instructions and declaring variables b) Getting input, storing values c) Displaying a message and displaying a value d) arithmetic expressions & operators (Introduction only)	<u>Exercises 02a, 02b, 02c</u>

7		<u>UNIT 3: 3 & UNIT 5.1</u>	Unit 3: 3.2 Pseudocode & Unit 5: 5.1 Pseudocode Instructions (3.2) a. pseudocode instructions (3.2): arithmetic expressions & operators and precedence	<u>Exercises 02d all</u> <u>Exercises 02e: #1, #2,</u> <u>#4b</u>
			b. sequence control structure (5.1) Grouping instructions together in a program using sequence control structure.	
8		<u>REVISION</u>	REVISION	<u>Additional Exercises</u>
9		<u>TEST 1</u>	Suggested Scope Unit 1, Unit 2, Unit 3 Part 1, Unit 3 Part 2 Environment/Context (Theory) General Problem Solving Writing simple individual pseudocode instructions, using arithmetic operators	
10		<u>UNIT 4: PART 1</u>	<u>Unit 4 Part 1 IPO Charts</u> a. IPO Charts	<u>Exercises 03: 3,4 Extra</u> <u>practice: Textbook:</u> <u>p48, p49, p54</u>
11		<u>UNIT 5: PART 1</u>	<u>Unit 5 Part 1 Simple Selection</u> a. Simple Selection control structure (IF else End If) (5.2.1) in pseudocode & relational operators	<u>Exercises 03:1,2,5,6,7</u> <u>Extra practice:</u> <u>Textbook: p. 48, p. 49,</u> <u>p. 54</u>
12		<u>UNIT 4: PART 2</u>	<u>Unit 4 Part 2 Flowcharts</u> a. Flowcharts incl. selection	<u>Exercises 04: 2, 5 Extra</u> <u>practice: Textbook p.57</u>
14		<u>UNIT 5: PART 2</u>	<u>Unit 5 Part 2 Compound Selection</u> a. Compound IF & Logical operators (5.2.2)	<u>Exercises 04: 1,3,4</u> <u>Extra practice:</u> <u>Textbook p.57</u>

13		<u>TEST 2</u>	Suggested Scope Unit 3 Part 3, Unit 4 & 6 Part 1, Unit 4 & 5 Part 2 Writing full pseudocode solutions IPO Charts, Flowcharts (not for them to draw) Using Simple Selection, using relational operators	
15		<u>UNIT 4:</u> <u>PART 3</u>	<u>Unit 4 Part 3 Trace Tables</u> a. Trace Tables ** rules * (4.3)	<u>Exercises 05 & 05b</u>

				<u>Extra practice: Text book p65, p70, p75</u>
16		<u>REVISION</u>	REVISION	
17		<u>UNIT 6</u>	<u>Unit 6: C # Programming</u> Introduce C #, concepts, IDE, syntax, etc. Practice writing C # programming on paper	<u>Prac 01, 02 on PCs</u> <u>Pracs 03, 04, 05 - Design on paper first</u>
18		<u>UNIT 5:</u> <u>PART 3</u>	<u>Unit 5 Part 3.1 FOR LOOP Iteration:</u> FOR (5.3) in pseudocode, trace table, how to accumulate, and how to count	<u>Exercises 06</u> <u>:1a,1b,2a,2b,4,5</u> <u>Extra practice:</u> <u>Textbook p126</u> <u>Prac 06A, 07 on PCs</u> <u>Design on paper first</u>
19		<u>REVISION</u>	REVISION	

20		<u>TEST 3</u>	Suggested Scope	
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			Unit 3 Part 3, Unit 4 Part 1 & 3, Unit 5 Part	
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			1, 2, 3.1	
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			Writing full pseudocode, C #	
--	--	--	------------------------------	--

			Simple and Compound Selection, FOR	
--	--	--	------------------------------------	--

			loop	
--	--	--	------	--

			Trace Tables	
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21		<u>UNIT 5:</u> <u>PART 4</u>	Unit 5.3.2: Iteration Control Structures <u>DO WHILE PRE-TEST LOOP</u> including with sentinel value in pseudocode, trace table, and C #	<u>Exercises 07</u> Extra practice: Textbook <u>p147</u> <u>Prac 06B - on PCs</u> <u>Design on paper first</u>
22		<u>REVISION</u>	REVISION	<u>Exercises 08</u> Extra practice: Textbook <u>p126, p147</u>
23		<u>UNIT 5:</u> <u>PART 5</u>	Unit 5.2.4: Advanced Selection Control Structures: <u>SELECT CASE control</u> <u>structures</u> in pseudocode, trace table, and C #	<u>Exercises 09</u> Extra practice: Textbook <u>p94, p103</u>
				<u>Prac 08A, 08B - on PCs,</u> <u>Design on paper first</u>
24		<u>UNIT 5:</u> <u>PART 5</u>	Unit 5.2.4: Advanced Selection Control Structures: <u>NESTED IF control structures</u> in pseudocode, trace table, and C #	<u>Exercises 09</u> Extra practice: Text book <u>p94, p103</u> <u>Prac 08A, 08B</u> <u>- on PCs</u> <u>Design on</u> <u>paper</u> <u>first</u>
25		<u>REVISION</u>	REVISION	<u>Exercises 10</u> <u>Additional revision</u> <u>exercises</u>
26		<u>TEST 4</u>	Suggested Scope Writing full pseudocode, C # IPO Charts and Trace Tables, Selection, Repetition - FOR and DO WHILE, SELECT CASE, NESTED IF	
27		<u>REVISION</u>	Whole year's work	
28		<u>REVISION</u>	Whole year's work	
29			PUBLICATION OF DP	
30			STUDY WEEK	

16. Notes on Plagiarism

The standard Harvard Referencing system is used at Walter Sisulu University – details are available on the University Intranet and from the Resource Centre.

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